Section 3.6

I. Implicit Differentiation

a. Graph of $F(x, y) = 0$
b. Vertical Line Test
c. Assume Local Restriction
d. Treat (locally) $y = f(x)$
e. Differentiate equation with respect x
f. Solve (the resulting linear equation) for $\dfrac{dy}{dx}$

II. Examples

a. $x^2 + y^3 = 64$

b. $x^3 y + x^2 y^2 = x^4 + y$

c. $\sin(xy) + x = e^{-x}$

III. Tangent Line Approximation to $F(x, y) = 0$ at $(x_{0,} y_0)$

a. $x^2 + y^2 = 25$ at $(3, -4)$

b. $x + xy^2 = x - 2y + 3$ at $(1,1)$

IV Higher Derivatives

a. $xy - y^2 = x + y$

V Derivatives of $y = \sqrt{x}$, $y = \sqrt[n]{x}$, $y = x^r$
$y = b^x$, $y = \ln x$, $y = \log_b x$

Examples

VI Derivative of Inverse Trigonometric Functions

a. $y = \sin^{-1} x$, $y = \tan^{-1} x$, $y = \sec^{-1} x$

Examples

VII Logarithmic Differentiation

a. Quotients, Products, Powers

$$y = \frac{x(x+1)^2}{(x-3)}, \quad y = \sqrt{x^3 + x}(3x^4 - x)^8$$

b. Exponentials

$$y = (x^2+1)^{x+1}, \quad y = x^{\cos x}$$